



COLUMBIA UNIVERSITY
IN THE CITY OF NEW YORK

STAT 5701

NUMERICAL SUMMARIES OF DATA

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NUMERICAL SUMMARIES OF DATA: CENTRAL TENDENCY

- **Exploratory Data Analysis (EDA):** The foundational phase of quantitative analysis requires establishing the expected value or central location of a dataset's distribution.
- **Distributional Location Parameters:** While individual observations within a numerical vector exhibit natural variance, metrics of central tendency mathematically estimate the focal point around which the underlying distribution converges.
- **Primary Estimators:** This analysis will evaluate two foundational statistical estimators for central tendency: the arithmetic mean and the median.

CENTRAL TENDENCY: MEAN AND MEDIAN

- **Concept:** In descriptive statistics, the arithmetic mean is a fundamental measure of central tendency that quantifies the expected value of a finite, discrete dataset.
- **Sample Representation:** Consider a sample of n independent observations drawn from a population. Let these individual numerical data points be defined by the sequence $x_1, x_2, \dots, x_n$.
- **Formulation:** The sample mean, universally denoted by the estimator $\bar{x}$, is computed as the aggregate sum of all observed values divided by the total sample size (n).
- This relationship is formally expressed by the following equation:

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

MEAN: THE FINE PRINT

- In data science and machine learning, the mean is frequently conceptualized not just as a simple calculation, but as the solution to an optimization problem. Specifically, the arithmetic mean is the value that minimizes the **Sum of Squared Errors (SSE)** or **Mean Squared Error (MSE)** for a given set of data.
- **The Optimization Objective:** Given a dataset of n observations $X = \{x_1, x_2, \dots, x_n\}$, we seek to find a constant scalar value c that represents the best single-point summary of the data. To do this, we define a cost function (or loss function), $J(c)$, using the squared L_2 norm (Euclidean distance):

$$J(c) = \sum_{i=1}^n (x_i - c)^2$$

- The mean, $\bar{x}$, is formally defined as the argument c that minimizes this cost function:

$$\bar{x} = \arg \min_c \sum_{i=1}^n (x_i - c)^2$$

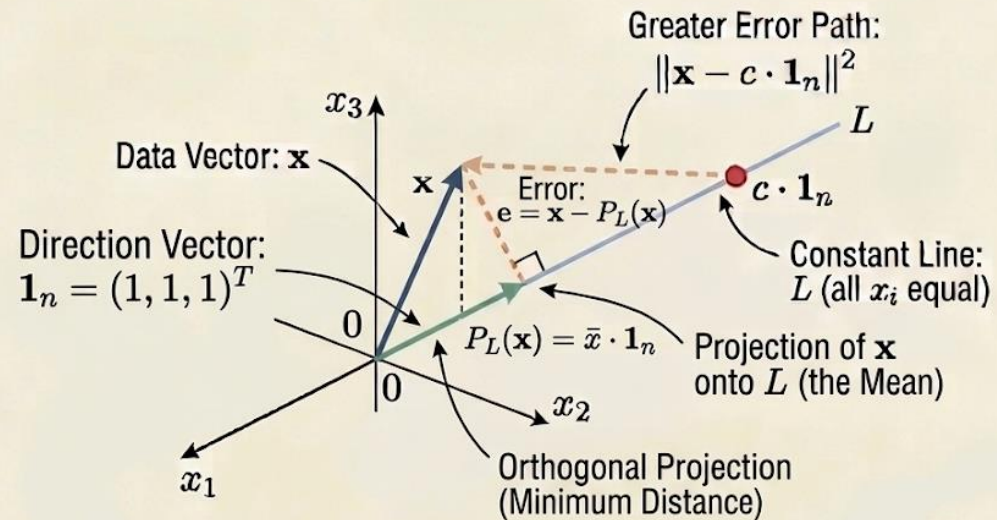
MEAN: THE VISUAL

The graph visualizes the arithmetic mean not merely as a formula, but as a spatial optimization problem. It illustrates how calculating the mean of a dataset

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{1}{n} \sum_{i=1}^n x_i$$

is equivalent to finding the shortest distance from a data point to a specific line in a high-dimensional space

OPTIMAL PROJECTION TO A CONSTANT LINE (1,1,1) DIRECTION



The mean $\bar{x}$ is the optimal scalar, and the mean vector is the orthogonal projection onto the “ones” direction.

$$\text{Optimal Scalar } \bar{x} = \operatorname{argmin}_c \|\mathbf{x} - c \cdot \mathbf{1}_n\|^2$$

$$\text{Geometric form: } P_L(\mathbf{x}) = \bar{x} \mathbf{1}_n = \left(\frac{\mathbf{x} \cdot \mathbf{1}_n}{\|\mathbf{1}_n\|^2} \right) \mathbf{1}_n \quad \text{Detailed Scalar: } \bar{x} = \frac{\mathbf{x} \cdot \mathbf{1}_n}{n} = \frac{\sum x_i}{n}$$

CENTRAL TENDENCY: MEAN AND MEDIAN

Sample Median Definition

- In statistics and data science, the **median** is a robust measure of central tendency representing the 50th percentile (0.5-quantile) of a probability distribution or sample.
- Given a sample of size n , let $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$ denote the **order statistics** (the observations sorted in non-decreasing order). The sample median, denoted as $\tilde{X}$ or $\text{Med}(X)$, is formally defined as:

$$\tilde{X} = \begin{cases} X_{(\frac{n+1}{2})}, & \text{if } n \text{ is odd} \\ \frac{X_{(\frac{n}{2})} + X_{(\frac{n}{2}+1)}}{2}, & \text{if } n \text{ is even} \end{cases}$$

MEDIAN: THE FINE PRINT

- **Robustness to Outliers:** The median has a breakdown point of **50%**, meaning up to half the data can be arbitrarily large or corrupted without shifting the median to infinity. This makes it preferable to the mean for skewed distributions.
- **L_1 Loss Minimizer:** The sample median is the point m that minimizes the sum of absolute errors (MAE):

$$\tilde{X} = \arg \min_{m \in \mathbb{R}} \sum_{i=1}^n |x_i - m|$$

- **Time to think:** Are there other estimators minimizing other losses?

L_2 VS L_1 : THE VISUAL COMPARISON

GEOMETRIC INTERPRETATION (UNIT SPHERE vs. UNIT OCTAHEDRON)

A set distance $R = 1$

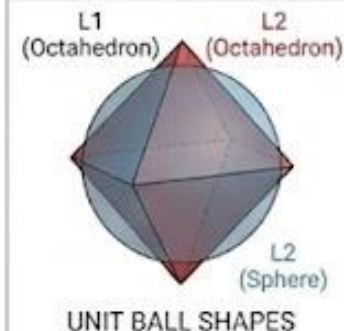
L_2 Norm (Euclidean, "Straight-Line"):

$$d_{L_2} = \sqrt{x^2 + y^2 + z^2} \leq R$$

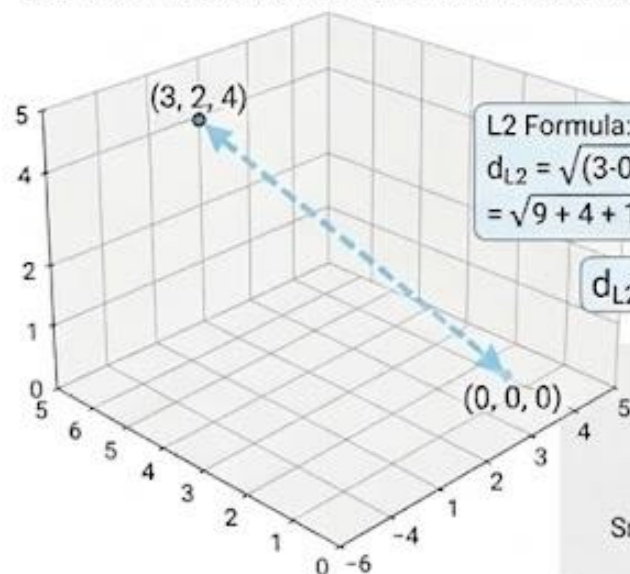
L_1 Norm (Manhattan, "Taxicab"):

$$d_{L_1} = |x| + |y| + |z| \leq R$$

UNIT BALL SHAPES



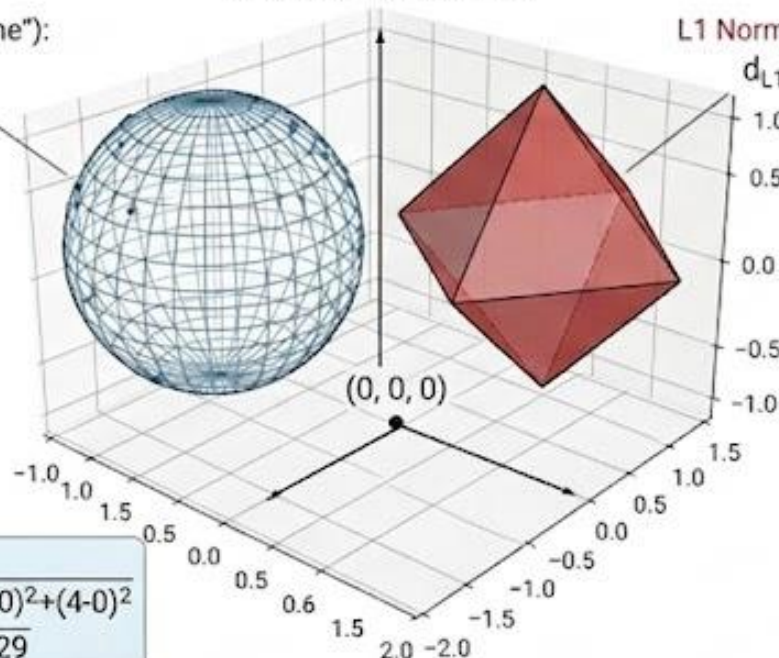
L_2 DISTANCE: STRAIGHT-LINE PATH.



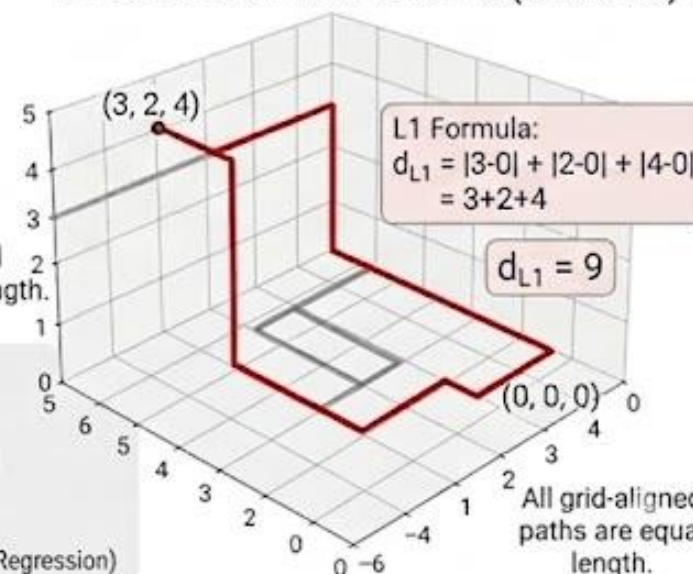
L_2 Formula:

$$d_{L_2} = \sqrt{(3-0)^2 + (2-0)^2 + (4-0)^2} \\ = \sqrt{9 + 4 + 16} = \sqrt{29}$$

$$d_{L_2} \approx 5.39$$



L_1 DISTANCE: GRID PATHS (TAXICAB).



L_1 Formula:

$$d_{L_1} = |3-0| + |2-0| + |4-0| \\ = 3 + 2 + 4$$

$$d_{L_1} = 9$$

KEY DIFFERENCES

L_2 (Euclidean)

Shortest physical path
Treats all directions equally
Smooth gradient (used in Ridge Regression)
Doesn't create sparsity

L_1 (Manhattan)

Path along orthogonal grid
Treats axes specially
Creates sharp corners
Encourages sparsity (used in Lasso Regression)

EXAMPLE 1: ILLUSTRATING THE DIFFERENCE

Suppose we have the following "data": 6, 3, 9

Compute **the mean** and median:

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{6 + 3 + 9}{3} = 6$$

Ordered data: 3, 6, 9

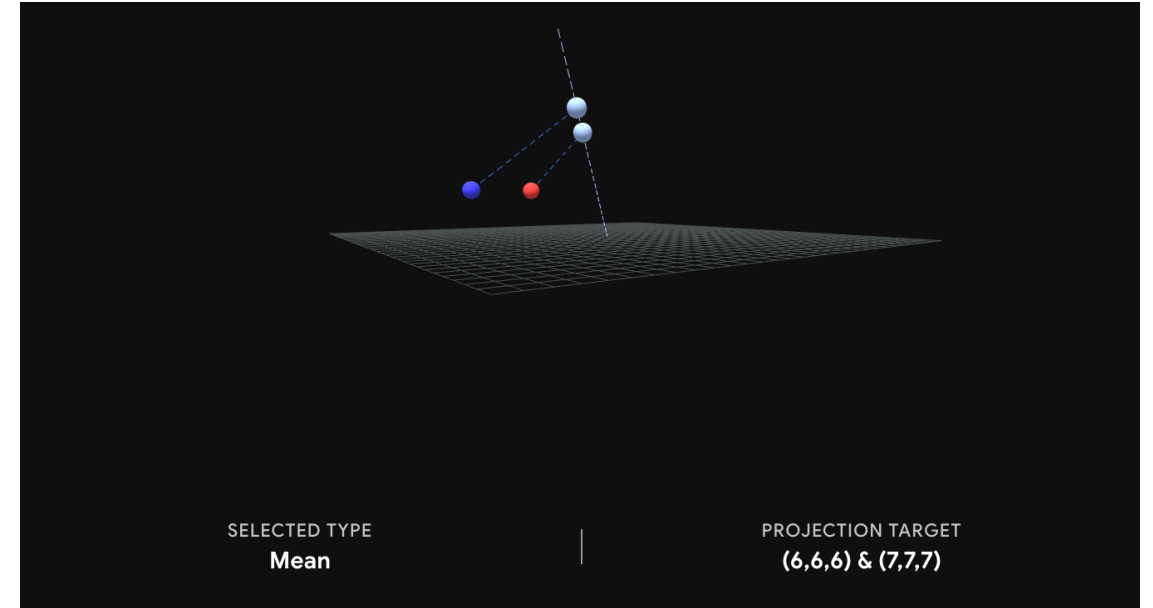
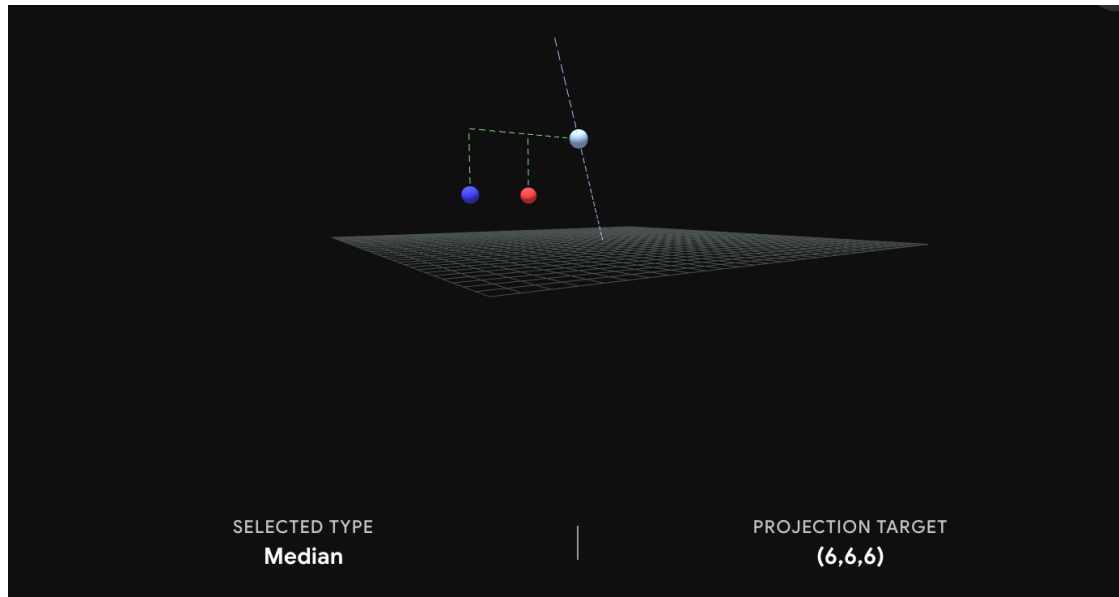
$$\tilde{x} = 6$$

What if the data were: 6, 3, 12

$$\bar{x} = \frac{6 + 3 + 12}{3} = 7$$

Ordered data: 3, 6, 12

$$\tilde{x} = 6$$



EXAMPLE 1: COMPARISON

CENTRAL TENDENCY: MEAN AND MEDIAN

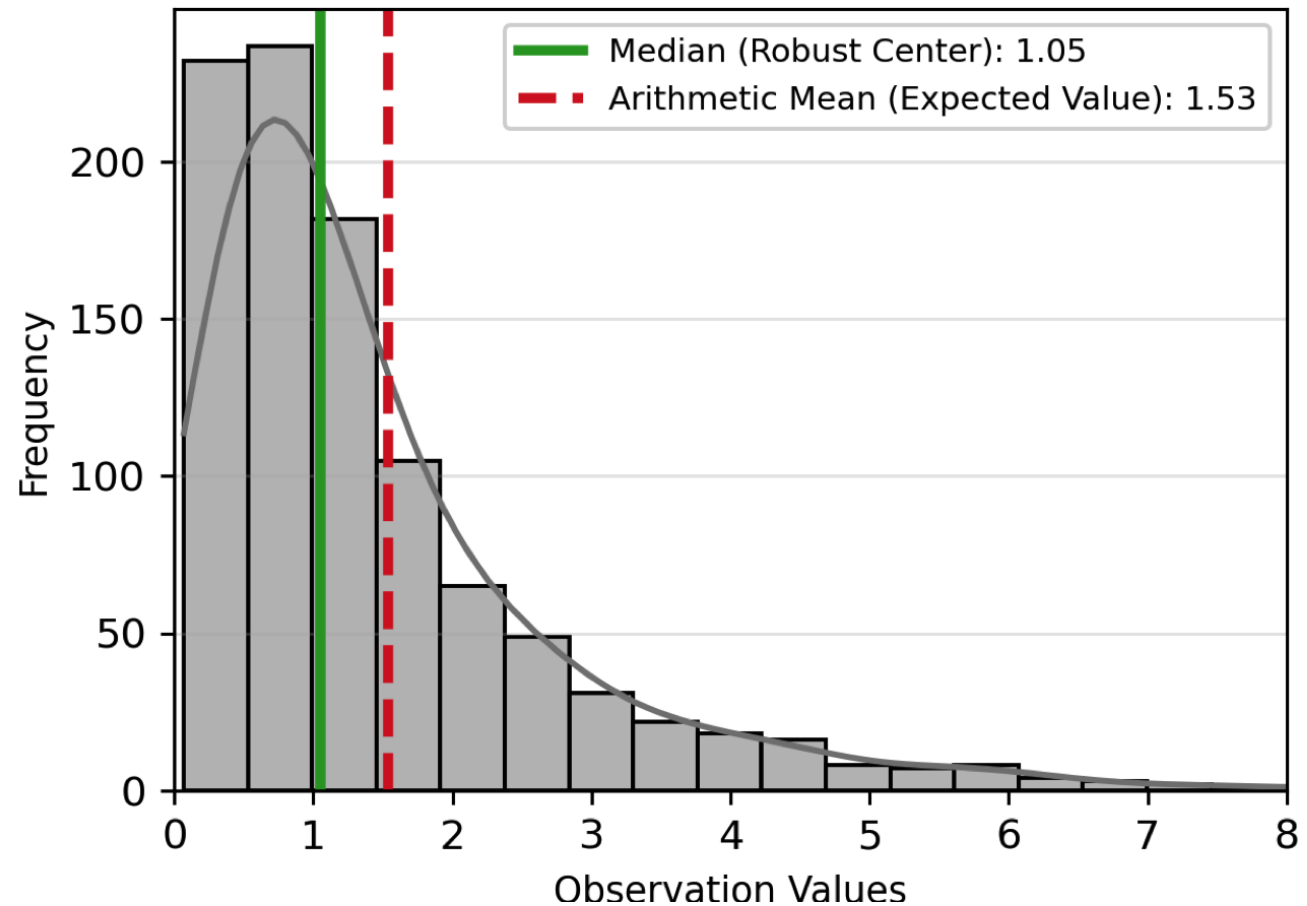
KEY IDEA:

- The **mean** is sensitive to extreme observations.
- The **median** is robust to extreme observations.

Question: How do we decide whether it is better to report the mean or the median as a measure of central tendency?

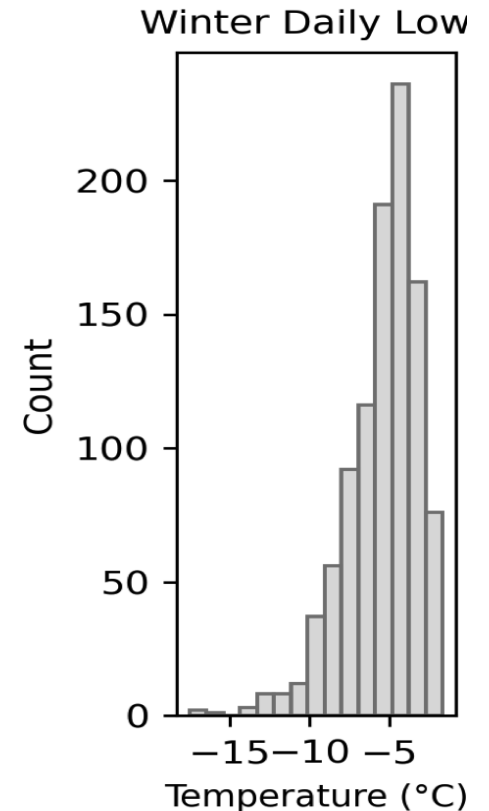
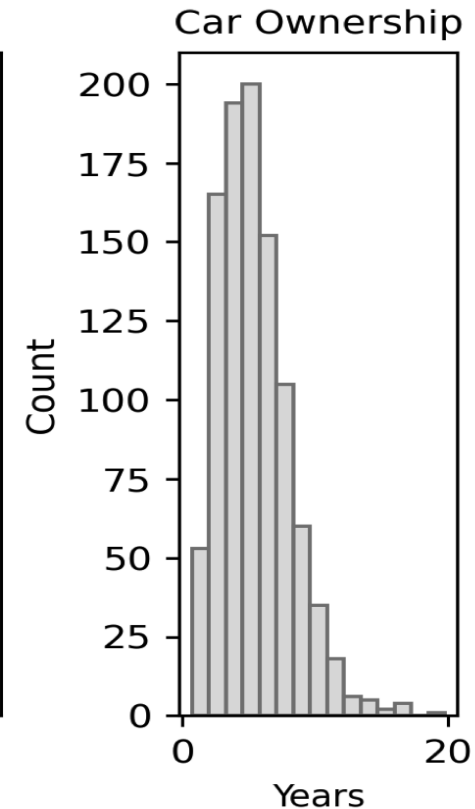
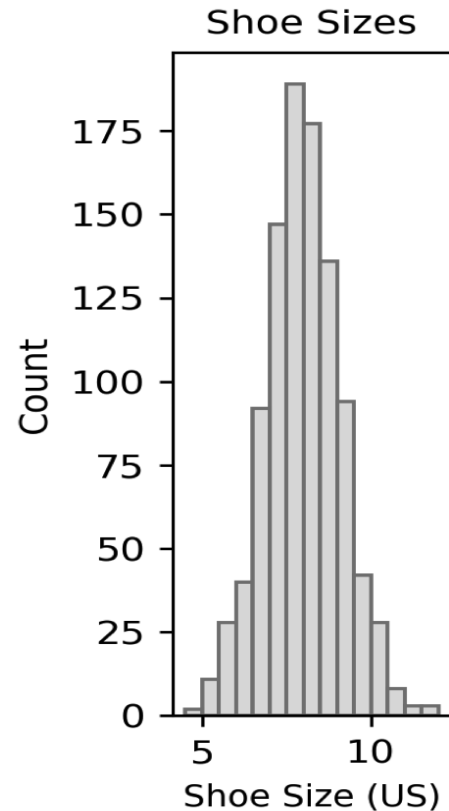
Answer: Generally, you want to choose the one that best represents a “typical” or “center” value in the data set.

Exploratory Data Analysis: Distributional Location Parameters



CENTRAL TENDENCY: MEAN AND MEDIAN

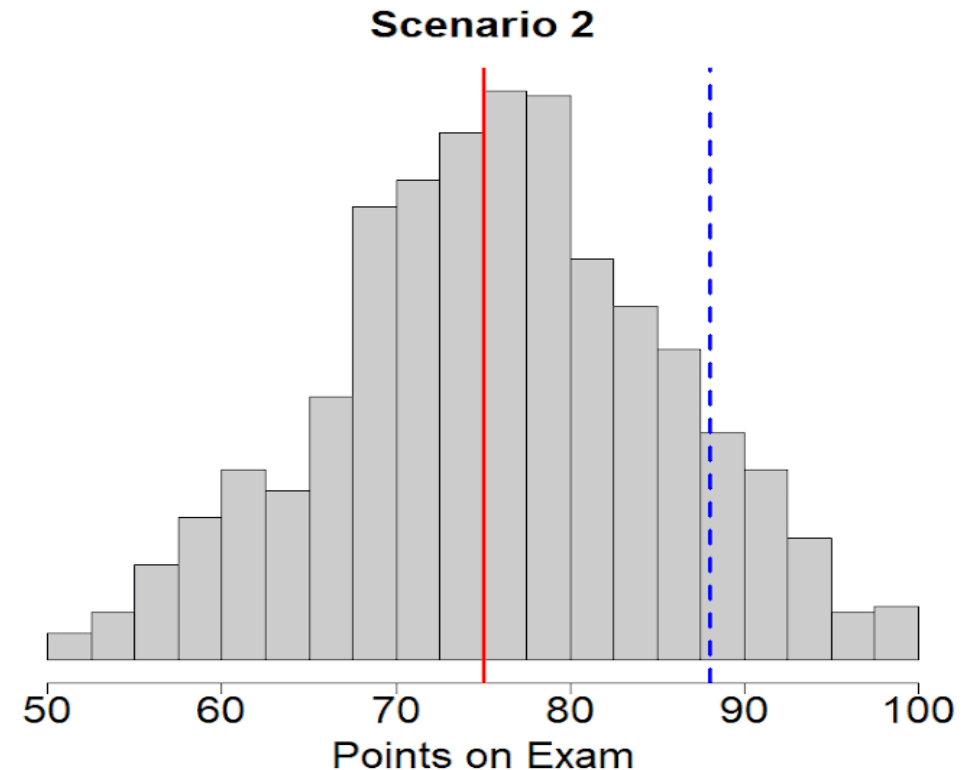
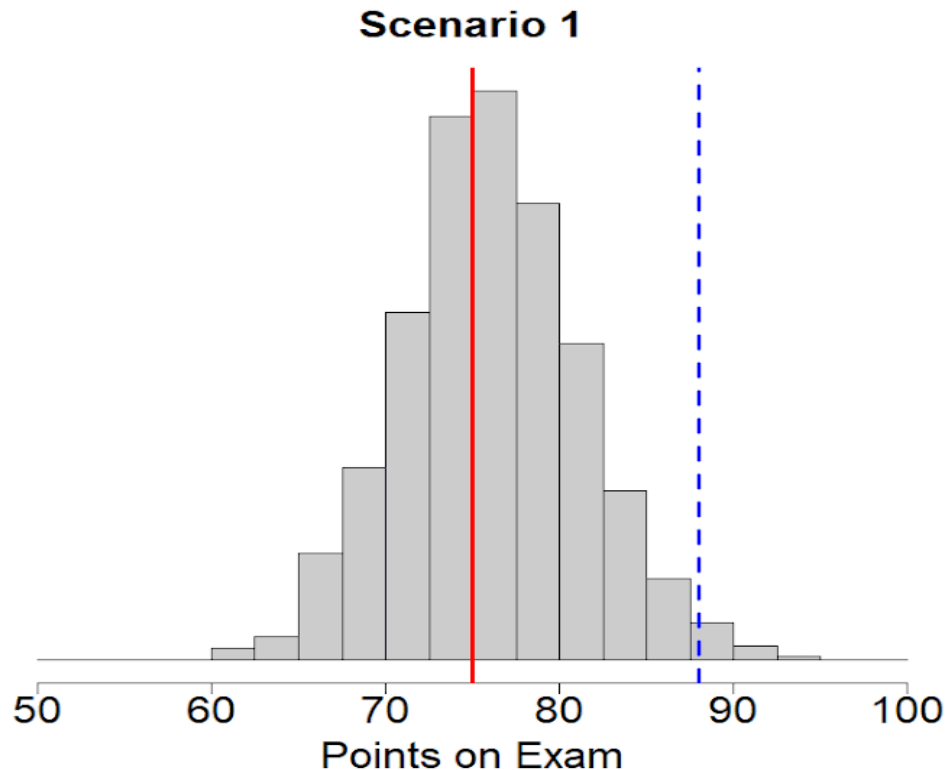
- Consider the following histograms.
- For each histogram, determine whether you would expect the mean and median values to be:
 - approximately equal,
 - for the mean $>$ median, or
 - for the mean $<$ median.



The bottom line: When analyzing histogram remember that the mean follows the skew of the histogram.

DESCRIBING VARIATION

Midterms are returned and “the average” was reported as 76 points out of 100 points. You received a score of 88 points? How do you feel about your performance in each scenario?



STATISTICAL DISPERSION AND VARIABILITY

- **Contextualizing Central Tendency:** Reporting measures of central tendency (e.g., mean, median) is statistically incomplete without a corresponding measure of dispersion. Dispersion metrics quantify the variance within a dataset, indicating the degree to which individual observations deviate from the central parameter.
- **Significance in Analysis:** Quantifying variability is a fundamental prerequisite for robust statistical inference, evaluating model performance, and accurately assessing uncertainty within a dataset.
- **Evaluating Metrics:** The following sections evaluate multiple statistical metrics of dispersion (e.g., variance, standard deviation, interquartile range). The selection of an optimal metric is context-dependent, requiring an assessment of each measure's mathematical properties, robustness to outliers, and inherent theoretical limitations.

RANGE

- Let X be a univariate sample consisting of n independent and identically distributed (i.i.d.) observations: $X = \{x_1, x_2, \dots, x_n\}$.
- If we sort these observations into order statistics such that $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$, where $x_{(1)}$ represents the sample minimum and $x_{(n)}$ represents the sample maximum, the sample range $R(X)$ is defined as:

$$R(X) = x_{(n)} - x_{(1)}$$

- In a continuous probability distribution with probability density function $f(x)$, the population range is defined by the support of the distribution—the interval $[a, b]$ where $f(x) > 0$.

The range exhibits several specific algebraic properties relevant to data transformations:

- **Non-negativity:** Because $x_{(n)} \geq x_{(1)}$ by definition, the range is strictly bounded at zero: $R(X) \geq 0$.
- **Translation Invariance:** Shifting a dataset by a constant c does not affect its dispersion.

$$R(X + c) = R(X)$$

- **Scale Equivariance:** Multiplying a dataset by a scalar c scales the range by the absolute magnitude of that scalar.

$$R(cX) = |c|R(X)$$

STATISTICAL LIMITATIONS: THE ROBUSTNESS PROBLEM

Zero Breakdown Point: The breakdown point of an estimator is the proportion of incorrect/extreme observations an estimator can handle before yielding an arbitrarily large result. The range has a breakdown point of exactly $\frac{1}{n}$. A single outlier—due to measurement error or anomalous behavior—will completely distort the range. To see why the range's "zero breakdown point" is such a critical concept in data science, consider a predictive maintenance model monitoring the operating temperature (in Celsius) of an industrial motor.

Distance-based machine learning algorithms (like k-Nearest Neighbors) and neural networks require inputs to be on a standardized scale. A common technique is **Min-Max Normalization**, which uses the range to map all data to a strict $[0,1]$ interval:

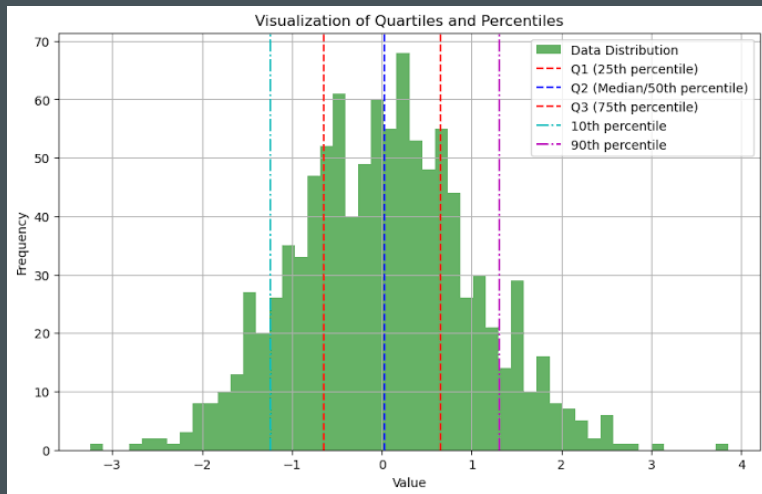
$$x'_i = \frac{x_i - x_{(1)}}{R(X)}$$

Suppose our IoT sensor records five normal operating temperatures and suffers a brief hardware fault, recording one extreme anomaly:

Because of that single anomalous reading, the sample range jumps from 1.7 to 60.5.

Observation (xi)	Sensor State	Scaled Value (xi')
44.8	Normal (Min)	0.000
45.2	Normal	0.007
45.9	Normal	0.018
46.1	Normal	0.021
46.5	Normal	0.028
105.3	Hardware Fault (Max)	1.000

PERCENTILES



- The **p-th percentile** is the value such that **p%** of the observations fall at or below that value.
- To find the *p*-th percentile empirically:
- **Sort** the dataset in ascending order so that $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$.
- **Calculate the rank index *i*** for the desired percentile. While there are several conventions, a standard baseline formula is:

$$i = \frac{p}{100} \times n$$

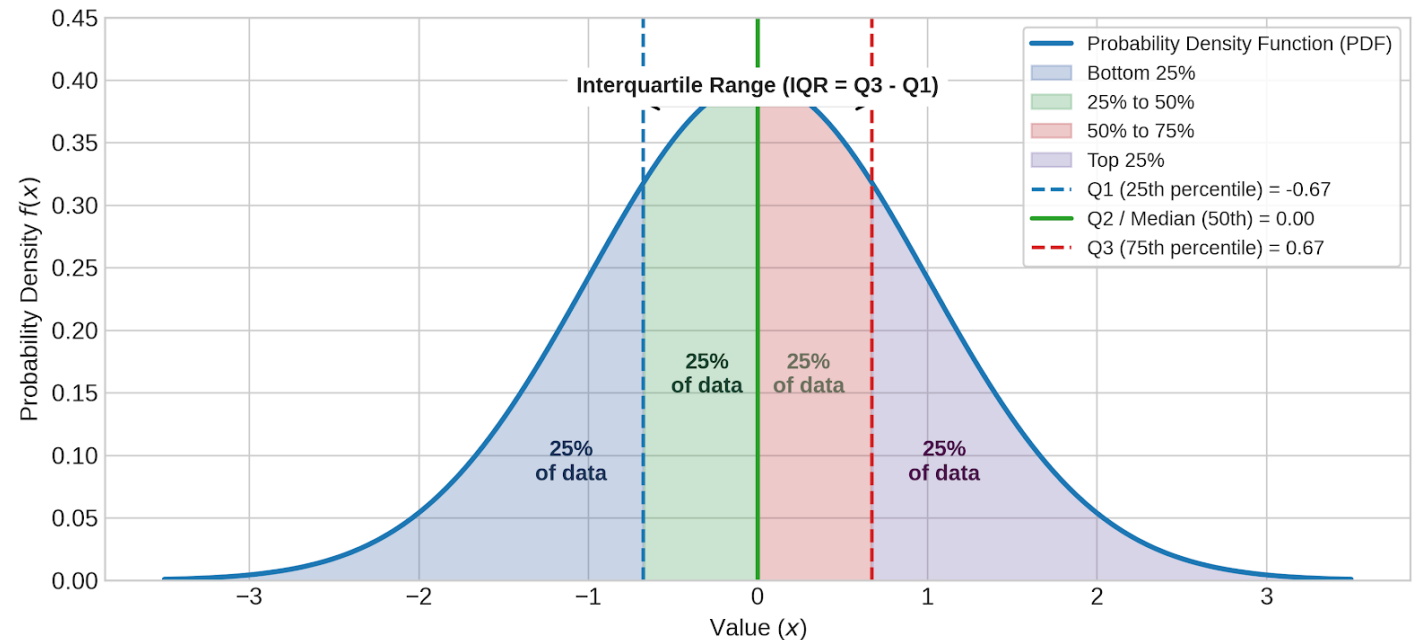
- **Interpolate:** If *i* is an integer, the percentile is the value at index *i*. If *i* is not a clean integer, statistical software (like Python's NumPy or pandas) typically uses linear interpolation between the two nearest ranks ($x_{\lfloor i \rfloor}$ and $x_{\lceil i \rceil}$) to estimate the exact value.

PERCENTILES

Order Statistics: Standard Quantiles

- **25th Percentile (Q_1 / First Quartile):** The value x at which the Cumulative Distribution Function (CDF) satisfies $P(X \leq x) = 0.25$. It partitions the lower 25% of the empirical distribution from the upper 75%, corresponding to the median of the lower half of the dataset.
- **50th Percentile (Q_2 / Median):** The value x at which the CDF satisfies $P(X \leq x) = 0.50$. It represents a non-parametric measure of central tendency that bisects the probability distribution into two equal halves, offering robustness against outliers.
- **75th Percentile (Q_3 / Third Quartile):** The value x at which the CDF satisfies $P(X \leq x) = 0.75$. It partitions the lower 75% of the empirical distribution from the upper 25%, corresponding to the median of the upper half of the dataset.

Visualizing Quantiles & Percentiles (Q_1, Q_2, Q_3) on a Continuous Distribution



IQR

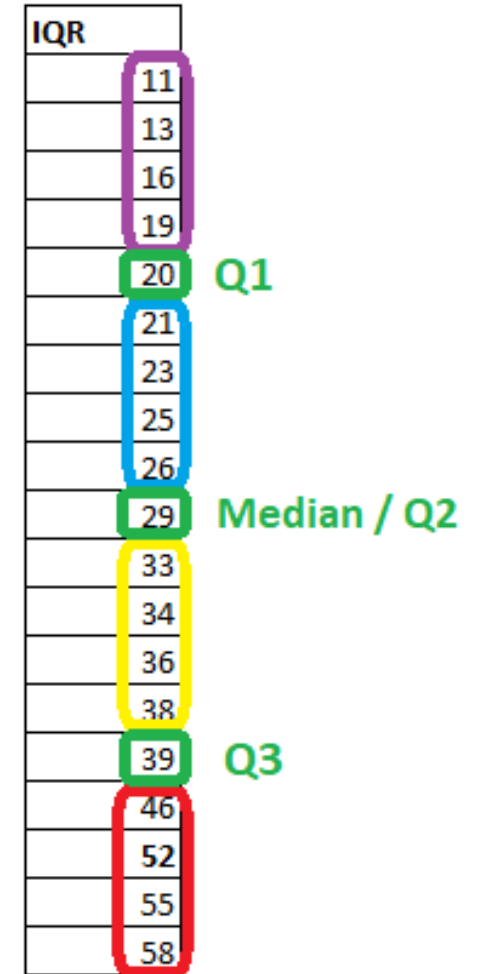
- In data science and descriptive statistics, the **Interquartile Range (IQR)** is a robust measure of statistical dispersion that quantifies the spread of the middle 50% of a dataset. By focusing only on the central portion of the data, the IQR provides a reliable metric for variability that is not heavily influenced by extreme values.

- The IQR is calculated by subtracting the first quartile from the third quartile of a rank-ordered dataset:

$$IQR = Q_3 - Q_1$$

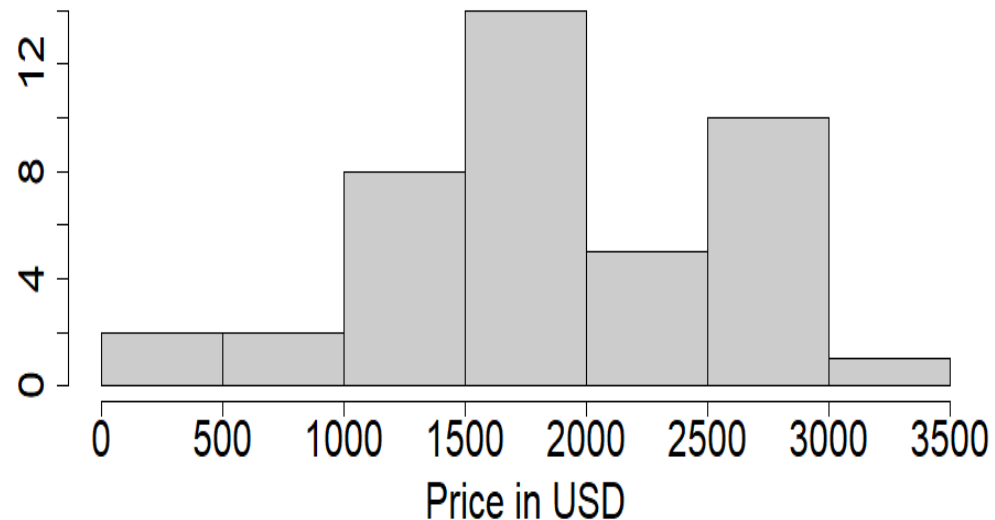
- Q_1 (First Quartile / 25th Percentile): The value separating the lowest 25% of the data from the highest 75%.

- Q_3 (Third Quartile / 75th Percentile): The value separating the lowest 75% of the data from the highest 25%.



EXERCISE: ROBUST STATISTICS

Now that we've discussed how to interpret histograms, let's see how they measure up against our statistical summaries. Below is a histogram of the sale price of 42 low-quality diamonds that are less than 1 carat in size. Their summary statistics are provided in the table next to it.



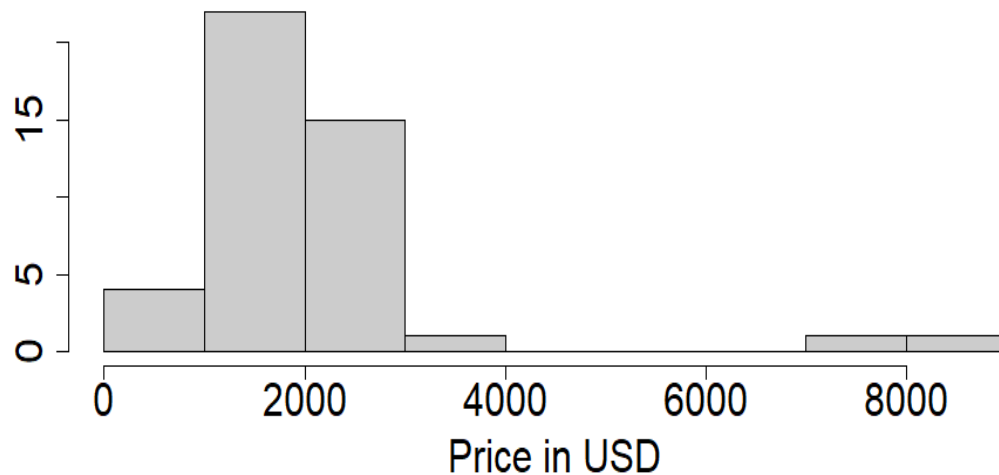
Statistic	Value
n	42
Min	\$416
Q1	\$1416
Median	\$1882
Mean	\$1924
Q3	\$2497
Max	\$3154
s	\$703

Compute the IQR and range of this data set:

IQR = 1081, Range = ? (finish as exercise)

EXERCISE: ROBUST STATISTICS

Suppose two new high-quality diamonds were accidentally mixed in with the original 42, one with sale price of \$7,393 and the other with a sale price of \$8,979. Compared to the low-quality diamonds, their sale prices are outliers. The summaries below show the new sale price distribution and its summary statistics.



Statistic	Value
n	44
Min	\$416
Q1	\$1429
Median	\$1899
Mean	\$2209
Q3	\$2537
Max	\$8979
s	\$1497

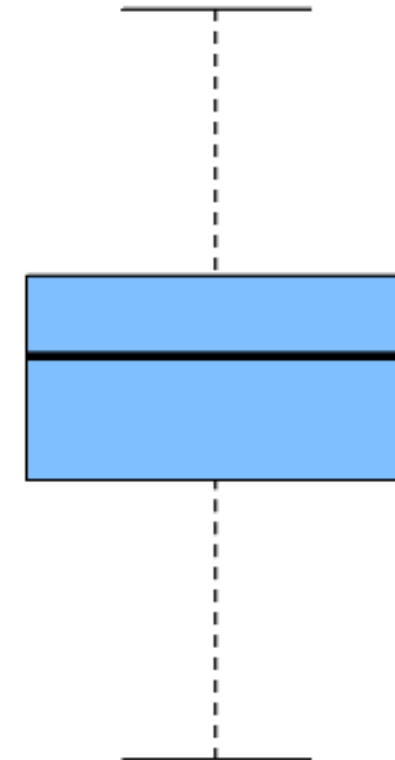
Compute the IQR and range of this data set. How do these summaries change after the high-quality diamonds are included?

ASIDE: VISUALIZING IQR WITH BOXPLOTS

A boxplot (or box-and-whisker plot) is a standardized, non-parametric visualization that displays the distribution of a dataset based on a five-number summary. Data scientists use it to quickly evaluate central tendency, dispersion, skewness, and anomalies without assuming an underlying normal distribution.

The Five-Number Summary

- **Minimum:** The lowest data point within the expected typical range, excluding statistical outliers.
- **First Quartile (Q_1):** The 25th percentile. Exactly 25% of the data falls below this threshold.
- **Median (Q_2):** The 50th percentile. The literal midpoint of the sorted dataset.
- **Third Quartile (Q_3):** The 75th percentile. Exactly 75% of the data falls below this threshold.
- **Maximum:** The highest data point within the expected typical range, excluding outliers.



VARIANCE & STANDARD DEVIATION

- Sample variance (s^2) is a measure of statistical dispersion that quantifies the average squared deviation of sample data points from their arithmetic mean ($\bar{x}$). In data science and statistics, it acts as an unbiased point estimator of the unknown population variance (σ^2).

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Notes:

- n : The sample size (total number of observations).
- $n - 1$ (**Bessel's Correction**): The adjusted degrees of freedom. Dividing by $n - 1$ rather than n compensates for using the sample mean $\bar{x}$ instead of the true population mean μ , eliminating negative bias in the variance estimate.
- The square root of sample variance yields the sample standard deviation ($s = \sqrt{s^2}$)

VARIANCE & STANDARD DEVIATION

Core Applications in Data Science

- **Feature Selection:** High variance indicates high information spread, whereas features with near-zero sample variance offer minimal discriminative power and are often dropped during preprocessing.
- **Dimensionality Reduction:** Linear techniques like Principal Component Analysis (PCA) compute eigenvectors along the directions of maximum sample variance to compress data without losing significant signal.
- **Bias-Variance Tradeoff:** Quantifies model sensitivity to training data fluctuations. High variance in a predictive model signals overfitting, where the model captures noise rather than underlying patterns.
- **Feature Scaling:** A critical parameter for calculating Z-score standardized values ($z = \frac{x - \bar{x}}{s}$) required by gradient-based and distance-based algorithms.

NOTATIONS FOR PARAMETERS VS STATISTICS

In order to indicate whether our summary values are calculated from population data and therefore are parameters or if they are from sample data and are statistics, we use special notation.

Measure	Parameter Notation	Statistic Notation
Mean	μ	$\bar{x}$
Variance	σ^2	s^2
Standard Deviation	σ	s
Proportion	p	$\hat{p}$

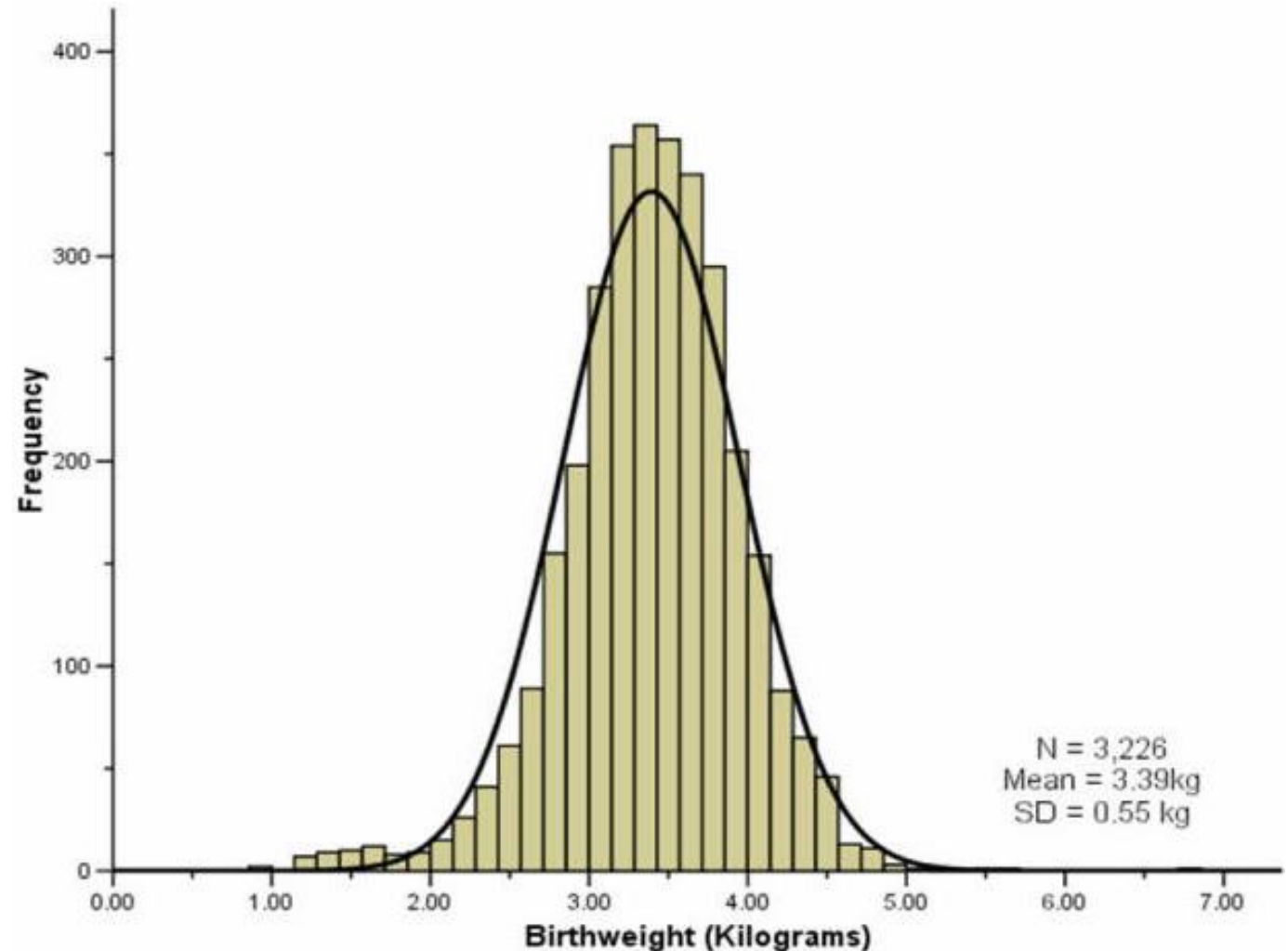
STANDARD DEVIATION

Notes about the standard deviation:

- **Dimensional Dispersion:** The sample standard deviation (s), derived as the square root of the variance (s^2), quantifies the dispersion of a dataset relative to its mean ($\bar{x}$) while preserving the original units of the data.
- **Zero Variance:** A standard deviation of $s = 0$ indicates a constant zero-variance feature, meaning all observations are identical and provide no informational entropy for modeling.
- **Non-Robustness:** Like the arithmetic mean, standard deviation is highly susceptible to skew from extreme outliers, making it a non-robust statistic compared to median-based measures like the interquartile range (IQR).
- **Applied Utility:** What do we do with the calculate value of s ?

STANDARD DEVIATION & THE EMPIRICAL RULE

- A symmetric bell-shaped pattern—known in data science as a **Normal Distribution** or **Gaussian Distribution**—occurs when most data points cluster around a central average, with extreme values tapering off equally on both the high and low ends.
- Here is what that looks like in the real world using infant birth weights.

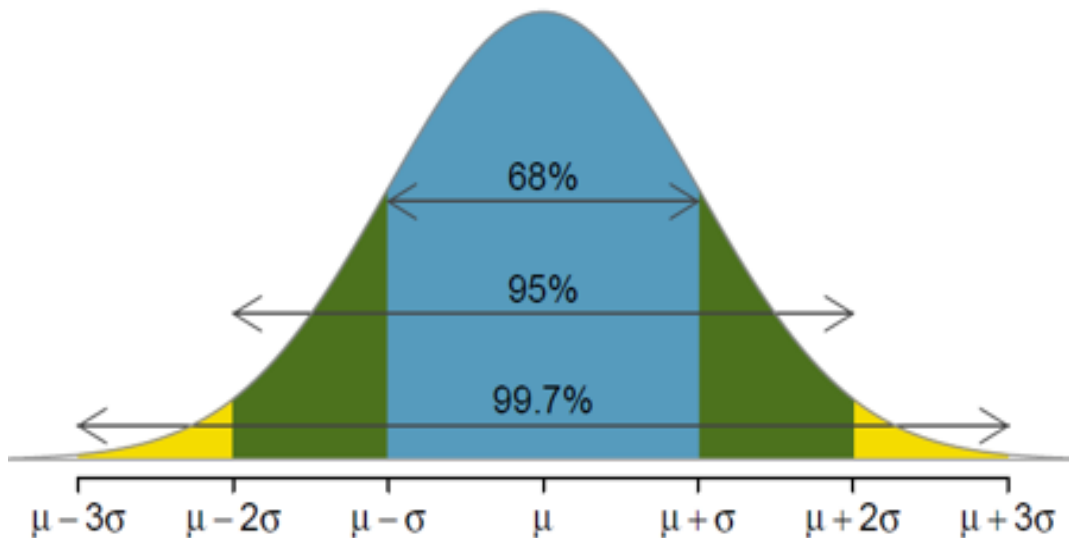


Distribution of birth weight in 3,226 newborn babies (data from O' Cathain et al 2002)

STANDARD DEVIATION & THE EMPIRICAL RULE

When our data is symmetric and bell-shaped,

Empirical Rule



- Approximately 68% of the data will be **within one standard deviation** of the mean.
- Approximately 95% of the data will be **within two standard deviation** of the mean.
- Approximately 99.7% of the data will be **within three standard deviation** of the mean.

EXAMPLE 2: STANDARD DEVIATION

Let's calculate the sample standard deviation of some heart rate data:

Resting Heart Rate	Deviation from $\bar{x}$	Squared Deviations
50	$50 - 65.5 = -15.5$	240.25
51	$51 - 65.5 = -14.5$	210.25
57	$57 - 65.5 = -8.5$	72.25
61	$61 - 65.5 = -4.5$	20.25
70	$70 - 65.5 = 4.5$	20.25
71	$71 - 65.5 = 5.5$	30.25
80	$80 - 65.5 = 14.5$	210.25
84	$84 - 65.5 = 18.5$	342.25
	Sum of squares:	1146

Sample variance: $s^2 = \frac{1146}{7} \approx 163.71$

Standard deviation, $s = \sqrt{163.71} \approx 12.8$

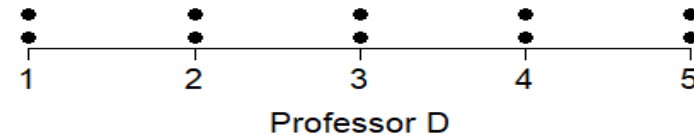
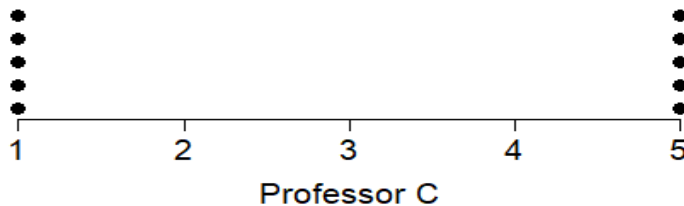
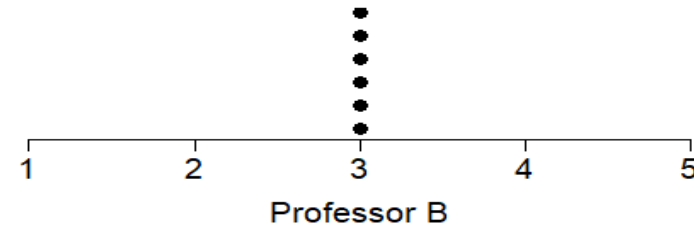
Interpretation of s: The resting heart rate of the

students in our sample are roughly
__12.8__ away

from the mean resting rate of
__65.5__, on average.

QUICK PRACTICE WITH THE STANDARD DEVIATION

At the end of each semester, responses from the Student-Instructional Rating System are provided to professors across the university. The dot plots below show student ratings (on a scale of 1-5) of four hypothetical professors (professors A – D). Arrange these professors in order from smallest variability in rankings to highest variability in rankings.



$$0 = \text{Var}(B) < \text{Var}(A) < \text{Var}(D) < \text{Var}(C)$$

QUICK PRACTICE WITH THE STANDARD DEVIATION

The table below asks you to compare the standard deviations of two data sets. Without doing any calculations. Choose one of the four statements below to describe the relationship between the data sets compared

Statement	Column A	Column B
	The standard deviation of {0.2, 0.4, 0.6, 0.8}	The standard deviation of {2, 4, 6, 8}
	The standard deviation of {1, 3, 5, 7, 9}	The standard deviation of {3, 5, 7, 9, 1}
	The standard deviation of {1, 3, 5, 7, 9}	The standard deviation of {1, 3, 5, 7, 9, 9}
	The standard deviation of {1, 3, 5, 7, 9}	The standard deviation of {1, 3, 5, 5, 7, 9}

- I. The quantity in column A is greater
- II. The quantity in column B is greater
- III. The two quantities are equal
- IV. The relationship cannot be determined from the given information